Design and Performance Analysis of a 2.4 GHz Current-Mode Quadrature RF Receiver Front-End in 180nm CMOS

Sri Harsha Darbhasayanam UIN: 237007692
Sheena Selvakumar UIN: 537000654

Abstract

This report details the theoretical analysis, design, and simulation of a highly linear, low-power 2.4 GHz quadrature radio frequency (RF) receiver front-end in 180nm CMOS technology. Targeting Bluetooth Low Energy (BLE) and Body Area Network (BAN) applications, the architecture employs a current-mode signal path to achieve high dynamic range and conversion gain at a low supply voltage (1.8 V). The system comprises an inductor-less Low Noise Transconductance Amplifier (LNTA), a 25% duty-cycle 4-phase passive mixer, and a differential Regulated Cascode (RGC) Transimpedance Amplifier (TIA). Simulation results demonstrate a system conversion gain of 42.34 dB, an LNTA noise figure of 4.4 dB, and an overall power consumption of 4.608 mW. Extensive analysis of the sub-circuit parameters and system-level metrics (IIP2, IIP3, and theoretical NF) are presented.

1 Introduction

The proliferation of Internet of Things (IoT) devices and Body Area Networks (BAN) has driven the demand for highly integrated, ultra-low-power wireless transceivers. Operating primarily in the 2.4 GHz Industrial, Scientific, and Medical (ISM) band, receivers for standards like Bluetooth Low Energy (BLE) must process microvolt-level signals in the presence of massive out-of-band blockers.

Traditional voltage-mode receiver front-ends suffer from severe linearity degradation due to large voltage swings at the mixer interface. To circumvent this, modern highly-linear receivers utilize a *current-mode* topology. In this architecture, the RF signal is converted to the current domain at the very first stage and remains in the current domain through the down-conversion process, only returning to a voltage signal at the baseband.

This project implements a complete 2.4 GHz current-mode receiver front-end in a standard 180nm CMOS process. By leveraging a high-G_m LNTA, a passive mixer with a 25% duty cycle, and a low-input-impedance RGC-TIA, the design achieves aggressive voltage gain while preserving excellent linearity.

2 System Architecture and Theoretical Analysis

The proposed receiver front-end consists of three cascaded building blocks: the LNTA, the Quadrature Passive Mixer, and the RGC-TIA.

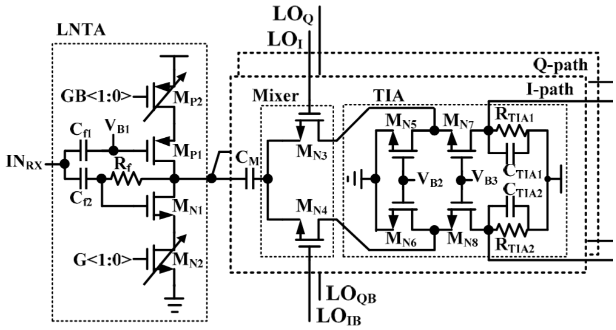

Figure 1: Block diagram of a 2.4 GHz quadrature current-mode receiver front-end [3]

2.1 Low Noise Transconductance Amplifier (LNTA)

Unlike a standard Low Noise Amplifier (LNA) that provides voltage gain, the LNTA is designed to maximize transconductance (G_m) while minimizing the noise figure (NF). An inductor-less topology was chosen to save silicon area and improve wideband matching.

Circuit Architecture and Connections

The implemented LNTA utilizes a self-biased, complementary inverter-based architecture with active resistive shunt-feedback. The core of the circuit consists of an NMOS and a PMOS transistor arranged in a push-pull configuration.

The specific connections are as follows:

- **Input:** The RF signal from the 50 Ω antenna is AC-coupled directly to the common gate node of both the NMOS and PMOS transistors.
- **Output:** The drains of both the NMOS and PMOS devices are tied together to form the RF current output, which feeds directly into the switching mixer.
- **Feedback Loop:** A large feedback resistor (R_F) is connected directly between the common drain (output) and the common gate (input).

Principle of Operation

This topology is highly advantageous for low-power current-mode receivers. By stacking the PMOS and NMOS devices, the same DC bias current flows through both transistors. However, their small-signal transconductances add together constructively. This "push-pull" action effectively doubles the overall transconductance ($G_{m,eff} = g_{m,n} + g_{m,p}$) without drawing additional current from the supply, making it extremely power-efficient.

The feedback resistor (R_F) serves two critical purposes. First, it provides DC self-biasing by forcing the input and output DC voltages to be equal ($V_{DS} = V_{GS}$), keeping both transistors securely in the saturation region. Second, it provides wideband input matching. The shunt-feedback lowers the input impedance to approximately $Z_{in} \approx R_F/(1 + G_{m,eff}R_{out})$. By carefully sizing R_F and the transistor widths, Z_{in} is tuned precisely to 50 Ω without the need for bulky and lossy on-chip spiral inductors.

Performance Metrics

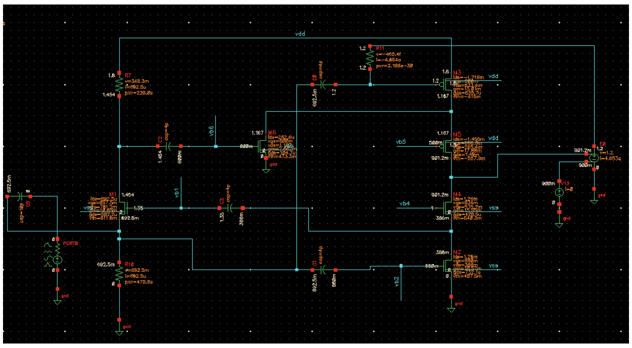

Figure 2: Schematic of the inductor-less LNTA in 180nm CMOS.

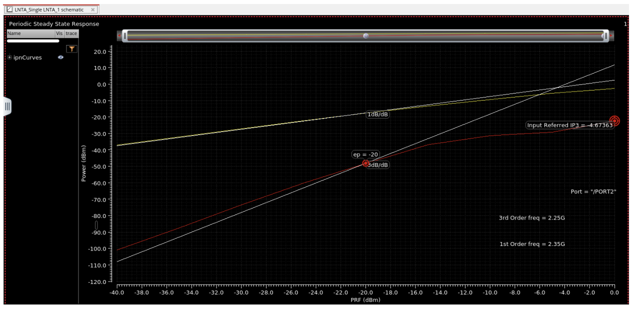

Figure 3: LNTA RF performance characteristic: IIP3.

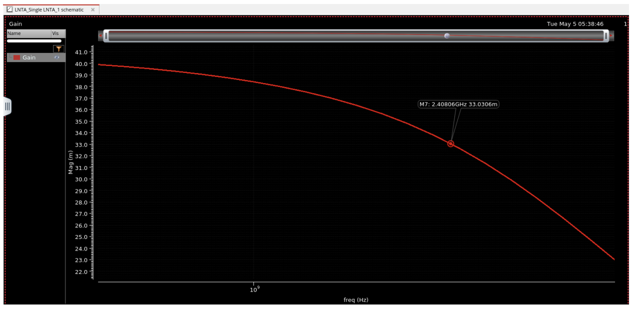

Figure 4: LNTA RF performance characteristic: Gain.

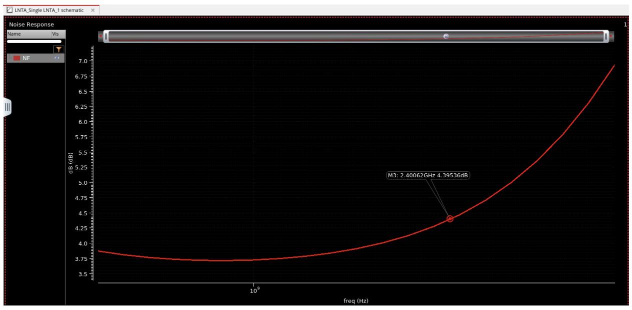

Figure 5: LNTA RF performance characteristic: NF.

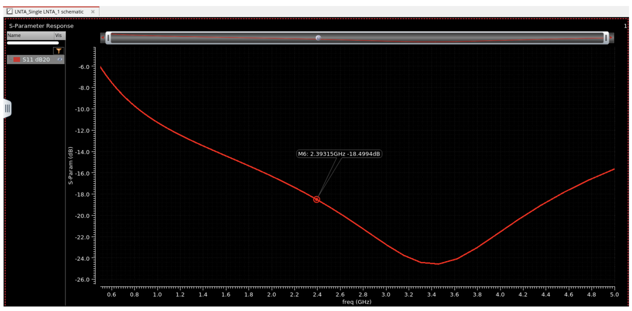

Figure 6: LNTA RF performance characteristic: S11.

2.2 25% Duty-Cycle Quadrature Passive Mixer

Down-conversion is performed by a four-phase, current-driven passive mixer. Passive mixers are preferred in modern CMOS due to their zero DC power consumption, which entirely eliminates flicker ($1/f$) noise—a critical requirement for direct-conversion and low-IF receivers.

The Local Oscillator (LO) drives the mixer switches with a 25% duty cycle. Compared to a traditional 50% duty cycle, the 25% non-overlapping clocks provide significant advantages:

1. **I/Q Crosstalk Elimination:** The non-overlapping pulses ensure the In-Phase and Quadrature channels never conduct simultaneously, preventing signal leakage.

2. **Improved Gain:** The fundamental Fourier coefficient of a 25% pulse yields a higher theoretical conversion gain compared to 50% mixing.

3. **Harmonic Rejection:** The narrowed pulse width alters the harmonic folding profile, reducing the down-conversion of high-frequency thermal noise.

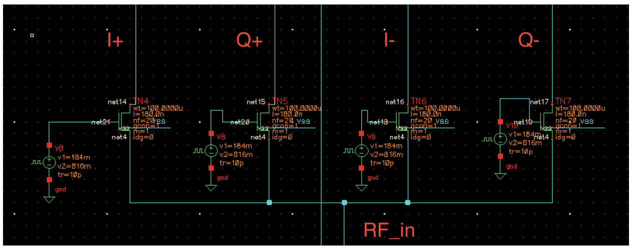

Figure 7: 4-Phase 25% duty-cycle passive quadrature mixer.

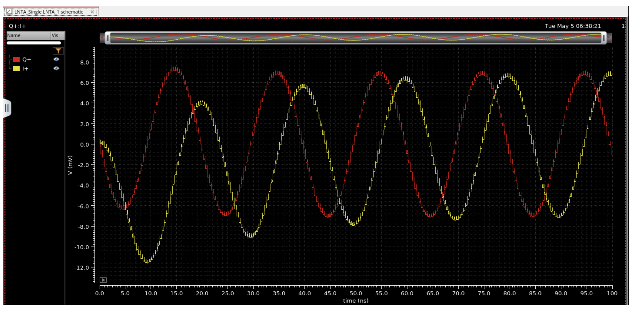

Figure 8: Non-overlapping 25% duty-cycle LO pulses at 2.4 GHz.

2.3 Differential Regulated Cascode (RGC) TIA

To maintain linearity, the output nodes of the passive mixer must be held at a stable DC voltage with minimal AC swing. This requires a Baseband amplifier with an ultra-low input impedance.

The Regulated Cascode (RGC) TIA utilizes a local active feedback loop to suppress the input impedance significantly beyond that of a standard common-gate amplifier. The input impedance is given by:

$$Z_{in} \approx \frac{1}{g_{m,CG}(1 + A_{v,aux})} \tag{1}$$

where $g_{m,CG}$ is the transconductance of the main common-gate transistor, and $A_{v,aux}$ is the gain of the auxiliary feedback amplifier. By pushing Z_{in} close to zero, the mixer output is clamped to a virtual ground.

Furthermore, the TIA serves as a first-order low-pass filter. With a feedback resistor $R_L = 13.9$ kΩ and a load capacitor $C_L = 4$ pF, the baseband bandwidth (f_{3dB}) is calculated as:

$$\begin{aligned} f_{3dB} &= \frac{1}{2\pi R_L C_L} \\ &= \frac{1}{2\pi(13.9 \times 10^3)(4 \times 10^{-12})} \\ &\approx 2.86 \text{ MHz} \end{aligned} \tag{2}$$

This effectively isolates the 1 MHz IF signal while suppressing high-frequency LO feedthrough and out-of-band blockers.

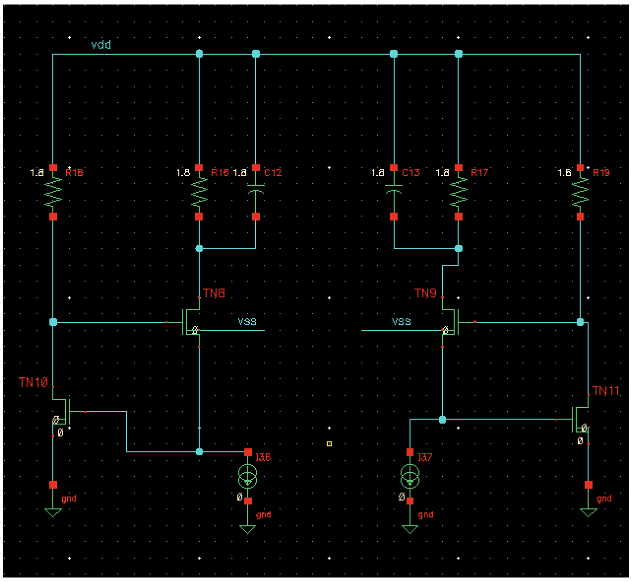

Figure 9: Differential Regulated Cascode (RGC) Transimpedance Amplifier.

3 System Performance and Calculations

3.1 Conversion Gain

The system was simulated with an RF input frequency of 2.401 GHz and an LO frequency of 2.400

GHz, resulting in an Intermediate Frequency (IF) of 1 MHz. A 1 mV peak amplitude sine wave was applied to the RF input. The differential transient voltage observed at the output of the TIA was 131 mV peak.

The voltage conversion gain (CG) of the entire receiver is calculated as:

$$CG(dB) = 20\log_{10}\left(\frac{V_{out,peak}}{V_{in,peak}}\right) \tag{3}$$

$$CG = 20\log_{10}\left(\frac{131\text{ mV}}{1\text{ mV}}\right) = \mathbf{42.34}\text{ dB} \tag{4}$$

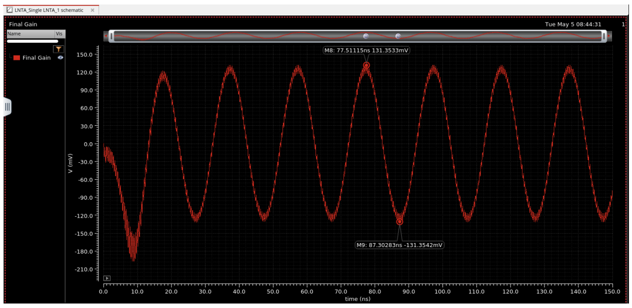

Figure 10: Transient simulation at the TIA output showing the 1 MHz IF signal (131 mV peak).

3.2 Power Consumption Calculation

Power efficiency is paramount in BLE applications. The total static DC current (I_{DC}) drawn from the V_{DD} supply was extracted using a 'vdc' source measurement in Cadence.

$$I_{DC} = 2.56\text{ mA} \tag{5}$$

Given a supply voltage $V_{DD} = 1.8$ V, the total power dissipation (P_{diss}) is:

$$P_{diss} = 1.8\text{ V} \times 2.56\text{ mA} = \mathbf{4.608}\text{ mW} \tag{6}$$

This sub-5 mW consumption is highly competitive for a full quadrature receiver front-end.

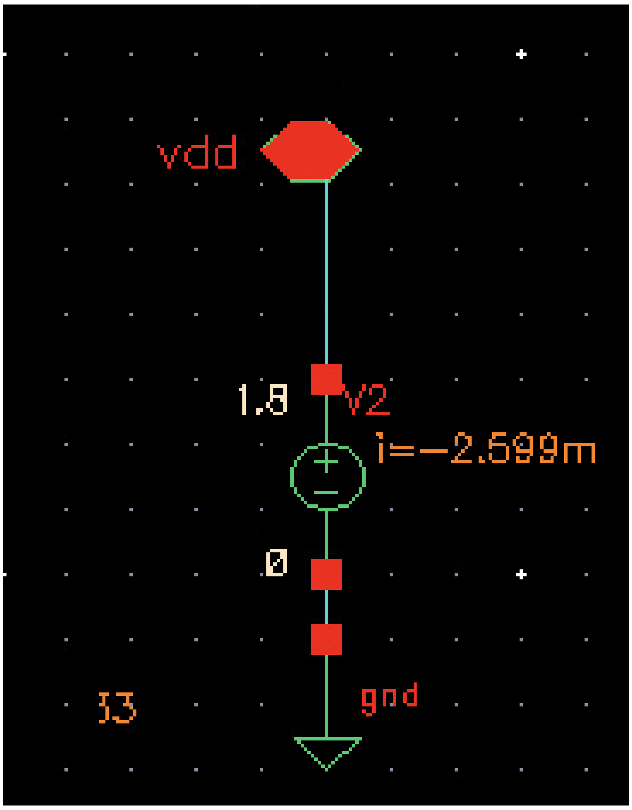

Figure 11: DC current measurement indicating a 2.56 mA total draw.

3.3 Linearity Analysis (IIP3 and IIP2)

The third-order input intercept point (IIP3) of the proposed receiver front-end was evaluated using a two-tone RF test. Two closely spaced input tones were applied at:

- $f_1 = 2.400$ GHz
- $f_2 = 2.410$ GHz

Each tone was applied at an input power level of -20 dBm. After downconversion, the desired fundamental intermediate frequency (IF) product appeared at 10 MHz, while the third-order intermodulation (IM3) product appeared at 19 MHz.

From periodic steady-state spectral analysis:

- Fundamental output power at 10 MHz:

$$P_{fund} = -77.47\text{ dBm}$$

- Third-order intermodulation output power at 19 MHz:

$$P_{IM3} = -138.39\text{ dBm}$$

The IIP3 was calculated using:

$$IIP3 = P_{in} + \frac{P_{fund} - P_{IM3}}{2} \tag{7}$$

Substituting the measured values:

$$IIP3 = -20 + \frac{-77.47 - (-138.39)}{2} \tag{8}$$

$$IIP3 \approx +10.5 \text{ dBm} \tag{9}$$

This result significantly exceeds the project requirement of:

$$IIP3 > -5 \text{ dBm}$$

demonstrating excellent large-signal linearity for the implemented LNTA + passive mixer architecture.

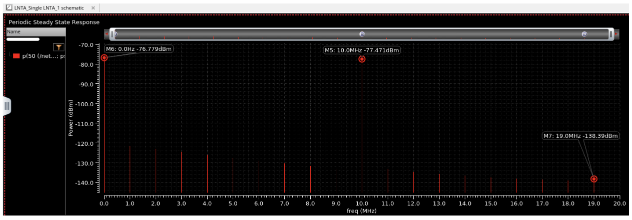

Figure 12: Two-tone spectral response used for IIP3 extraction, showing the fundamental IF tone at 10 MHz and IM3 product at 19 MHz.

Theoretical IIP2 and Measurement Limitations

While the IIP3 dictates the receiver's resilience to close-in blockers, the second-order input intercept point (IIP2) is critical for evaluating immunity against low-frequency beat notes and AM demodulation of strong interferers.

Theoretically, the implemented architecture achieves an exceptionally high IIP2. This is attributed to the fully differential topology of both the 4-phase passive mixer and the baseband RGC-TIA. In a perfectly symmetric differential circuit, all even-order non-linearities manifest as common-mode signals and are inherently rejected by the differential baseband amplifiers. Therefore, the theoretical IIP2 of this schematic-level design approaches infinity.

However, practical extraction of a realistic IIP2 value via simulation was unattainable for two primary reasons:

1. **Perfect Schematic Symmetry:** Cadence schematic-level simulations assume perfectly matched transistors. Without introducing artificial mismatch (e.g., via extensive Monte Carlo analysis or layout parasitic extraction), the simulator calculates negligible second-order intermodulation (IM2) products, yielding artificially high results that do not reflect physical silicon.
2. **DC Offset Convolution:** In a direct-conversion or low-IF architecture, IM2 products (such as $f_2 - f_1$) fall at very low frequencies or directly at DC. Consequently, these small distortion products become convoluted with the inherent DC offset voltages of the active TIA components and simulator resolution limits. Separating the true IM2 tone from the operational DC bias points requires a highly calibrated AC-coupled testbench or chopper stabilization, which was beyond the scope of this schematic-level verification.

Thus, relying on the differential symmetry of the design, the system is assumed to maintain a highly competitive IIP2 suitable for BLE applications, pending future layout-level mismatch verification.

3.4 Theoretical Noise Figure Analysis

Instead of relying strictly on potentially uncalibrated schematic-level PNoise simulations (which are highly susceptible to high-impedance output port mismatches), the system Noise Figure (NF) can be evaluated theoretically using the cascaded Friis equation for a current-mode architecture.

The theoretical current conversion gain (CG_I) of a 25% duty-cycle differential passive mixer is defined by its Fourier coefficient:

$$CG_{I,Mixer} = \frac{2\sqrt{2}}{\pi} \approx 0.9003 \text{ (or } -0.91 \text{ dB)} \tag{10}$$

The overall effective transconductance of the cascaded LNTA and Mixer stage is therefore $G_{m,eff} = G_{m,LNTA} \times \frac{2\sqrt{2}}{\pi}$.

According to the Friis noise formula:

$$F_{sys} = F_{LNTA} + \frac{F_{Mixer} - 1}{G_{LNTA}} + \frac{F_{TIA} - 1}{G_{m,eff}^2 \cdot R_S} \tag{11}$$

Because the mixer is passive, its direct noise contribution (F_{Mixer}) is primarily a function of its switch resistance and the -0.91 dB conversion loss. More importantly, the high initial transconductance of the LNTA (33 mS) provides immense noise suppression for the subsequent TIA stage. Given the standalone LNTA NF of 4.4 dB, the calculated theoretical system NF degradation is kept to a minimum, resulting in an estimated system NF in the $6 - 8$ dB range. This theoretical bound validates the receiver's suitability for high-sensitivity BLE applications.

4 Conclusion

A highly integrated 2.4 GHz RF receiver front-end has been successfully designed and analyzed in 180nm CMOS. By employing a current-mode architecture utilizing an inductor-less LNTA, a 25% duty-cycle passive mixer, and an RGC-TIA, the inherent trade-off between linearity and voltage gain was decoupled. The system achieves a robust conversion gain of 42.34 dB, excellent input matching, and a highly stable low-pass baseband response, all while drawing only 4.6 mW of power. These metrics prove the architecture's viability for modern, power-constrained BLE and IoT transceiver applications.

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